

Clinical Staff Feedback Form

Automated PAR Index Calculation System

All responses are strictly confidential

Purpose: This form collects your expert clinical opinion on the usability, accuracy, and practical value of our web-based system for automated PAR Index calculation. Your input will directly inform our development priorities for Semester 4 and shape the system's readiness for real-world clinical deployment.

Dr. D.D. Vithanachchi

1 Respondent Information

Clinical Role (~~Consultant Orthodontist~~ / ~~Registrar~~ / ~~Dental Surgeon~~ / ~~Clinical Researcher~~ / ~~Academic~~ / ~~Other~~)

Years of Clinical Experience (< 2 / 2–5 / 6–10 / 11–20 / > 20)

Institution / Hospital

Fac. of Dental Sciences

Email Address (optional — for follow-up only)

Have you used any similar digital orthodontic assessment tools before? Circle one:

☐ Yes

☒ No

☐ Unsure

If yes, please name the system(s):

2 General Usability

Rate your level of agreement with each statement. Tick one box per row.

1 = Strongly Disagree · 2 = Disagree · 3 = Neutral · 4 = Agree · 5 = Strongly Agree

Statement	1 Str. Disagree	2 Disagree	3 Neutral	4 Agree	5 Str. Agree
The overall interface is intuitive and easy to navigate.	☐	☐	☐	☐	☒
I was able to understand the system's output without additional guidance.	☐	☐	☐	☐	☒
The workflow matches how I approach PAR scoring in clinical practice.	☐	☐	☐	☐	☒
The system minimises unnecessary complexity for clinical staff.	☐	☐	☐	☐	☒
I would feel comfortable using this system with minimal prior training.	☐	☐	☐	☐	☒

3 Feature-Specific Evaluation

Rate the quality / usefulness of each system feature. Tick one box per row.

1 = Very Poor · 2 = Poor · 3 = Fair · 4 = Good · 5 = Excellent

Statement	1 Very Poor	2 Poor	3 Fair	4 Good	5 Excellent
File Upload — ease of uploading 3D models (.STL / .OBJ) and file validation feedback	☐	☐	☐	☐	☒
Automated Landmark Detection — accuracy and reliability of AI-identified anatomical points	☐	☒	☒	☐	☐
3D Visualisation Viewer — rendering quality, zoom / rotate / pan responsiveness	☐	☐	☐	☐	☒
Manual Override Tool — ability to adjust and correct AI-placed landmark positions	☐	☐	☐	☐	☒
PDF Report Generation — clarity, completeness, and clinical usefulness of the output report	☐	☐	☐	☒	☐

4 Clinical Assessment & Confidence

4.1 How easy was it to upload a 3D dental model and initiate PAR calculation? Tick one.

☐ Very Difficult ☐ Difficult ☐ Neutral ☐ Easy ☒ Very Easy

4.2 How would you rate the 3D model viewer's performance and visual clarity? Tick one.

☐ Very Poor ☐ Poor ☐ Acceptable ☐ Good ☒ Excellent

4.3 How useful is the manual landmark override feature for clinical practice? Tick one.

☐ Not useful at all ☐ Slightly useful ☐ Moderately useful
☐ Very useful ☒ Essential

4.4 How suitable are the generated PDF reports for inclusion in clinical records? Tick one.

☐ Not suitable ☐ Needs major revision ☐ Needs minor revision
☐ Mostly suitable ☐ Fully suitable

4.5 How confident are you in the clinical accuracy of the AI-computed PAR scores? Tick one box.

1 = Not at all confident · 10 = Extremely confident

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☒ 7 ☐ 8 ☐ 9 ☐ 10
Not at all confident Extremely confident

4.6 What concerns, if any, do you have about integrating AI-based PAR scoring into clinical workflows? Tick all that apply.

- ☒ Accuracy and reliability of AI landmark detection ☐ Liability and accountability for AI-generated results
☒ Data privacy and patient confidentiality ☒ Over-reliance on automation reducing clinical skill
☐ System errors causing incorrect treatment decisions ☐ Resistance or lack of trust from the clinical team
☐ No major concerns

5 Efficiency & Time Savings

5.1 Compared to manual PAR scoring, how much time do you estimate this system saves per case? Tick one.

- ☐ No time saved ☐ Less than 5 minutes ☐ 5–10 minutes
☒ 10–15 minutes ☐ More than 15 minutes

5.2 Which stage of the digital workflow felt the most time-consuming? Tick one.

- ☐ File Upload ☐ AI Landmark Detection ☒ Manual Correction / Override
☐ Report Export ☐ None — all stages were fast

6 Clinical Validation & Error Analysis

6.1 In approximately what percentage of cases did you feel the need to use the Manual Override tool? Tick one.

- ☐ 0% — AI was always accurate ☐ Less than 25% ☐ Around 50%
☐ More than 75% ☒ Used override on every case

6.2 Are there specific dental landmarks where the AI consistently underperforms? Tick all that apply.

- ☐ Upper anterior segment / overjet ☐ Lower anterior segment
☐ Right buccal occlusion (molar relationship) ☐ Left buccal occlusion (molar relationship)
☐ Overbite / vertical overlap ☐ Centreline / midline deviation
☐ No consistent underperformance noticed

6.3 On a scale of 1–10, how closely did the AI-generated PAR score match your own manual calculation for the same model? Tick one box.

1 = Completely different · 10 = Perfect match

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10
Completely different Perfect match

7 Technical & System Performance

7.1 How would you rate the system's responsiveness when manipulating the 3D model (zoom, rotate, pan)? Tick one.

- ☐ Very laggy / unusable ☐ Noticeable lag ☐ Acceptable
☐ Smooth ☒ Very smooth / responsive

7.2 If a file upload failed or a calculation crashed, was the error feedback from the system helpful enough to resolve the issue? Tick one.

- ☐ Did not encounter any errors ☐ Error message was confusing ☐ Partially helpful
☐ Mostly helpful ☐ Fully helpful — I resolved it immediately

8 Integration & Compatibility

8.1 Did you experience any lag or rendering issues on the device you were using? Tick one, then specify device below.

- ☒ No issues at all ☐ Minor lag only ☐ Moderate lag
☐ Significant lag ☐ System was unusable on my device

Device used: **Tick one.**

- ☐ Laptop ☐ Desktop PC ☐ Tablet ☐ Other (specify below)

If 'Other', specify device: _____

8.2 How important is it for this system to sync directly with your existing Patient Management System (PMS)? Tick one.

- ☐ Not important ☐ Slightly important ☐ Moderately important
☒ Very important ☐ Essential — I would not adopt it without this

9 PDF Report Utility

9.1 Does the PDF report provide enough visual evidence (e.g. landmark screenshots) to justify the PAR score to a third party or auditor? Tick one.

- ☐ Definitely not ☐ Probably not ☐ Unsure ☐ Probably yes ☐ Definitely yes

9.2 How effective would this report be as a visual aid when explaining treatment needs to a patient? Tick one.

- ☐ Not effective at all ☐ Slightly effective ☐ Moderately effective
☐ Very effective ☒ Extremely effective

10 Overall Assessment

10.1 Overall, how would you rate this system for clinical use? Tick one.

- ☐ Very Poor ☐ Poor ☐ Fair ☒ Good ☐ Excellent

10.2 Would you recommend this system to colleagues or other orthodontic institutions? Tick one.

- ☐ Definitely not ☐ Probably not ☐ Unsure ☐ Probably yes ☒ Definitely yes

10.3 Which features do you consider most valuable for clinical practice? Tick all that apply.

- ☒ Automated landmark detection ☒ 3D model visualisation
☒ Manual override and correction ☒ Longitudinal patient tracking
☒ Downloadable PDF reports ☐ Model versioning and reproducibility
☐ Multi-institution / cloud access

11.1 What specific improvements would most increase your confidence in using this system clinically?

If we can develop fully automated landmark identification that would be very valuable.

11.2 Are there any clinical or institutional barriers that might prevent adoption of this system at your facility? (e.g. IT infrastructure, ethical clearance, data governance, training requirements)

No, If any thing arise we can take proper approval for it.

11.3 If you could automate one additional part of your clinical workflow (e.g. cephalometric analysis, treatment outcome prediction), what would be your top priority?

Cephalometric analysis

11.4 Any additional comments, suggestions, or features you would like to see in future versions?

As I already mention making the landmark identification fully automated is the key feature to success in this software. Thank

By completing and returning this form, you consent to your anonymized responses being used solely for the research and development of the Automated PAR Index System at the University of Peradeniya. Contact e22056@eng.pdn.ac.lk.